Putter Shot Dynamics of Shiftless Collatz on a Base-3 Logarithmic Spiral Green:

The Interplay Between Ball Speed (3n) and the Grain of the Green (LSB) Approaching Infinite Cups

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1. Introduction

This report presents a geometric visualization of the Shiftless Collatz model, which eliminates the information loss caused by the division (right-shift) step in the classical Collatz map. The model is defined as:

$$n_{k+1} = 3n_k + \mathrm{LSB}(n_k)$$

Here, LSB (lowest significant bit) denotes the least significant bit of n_k in its binary representation.

By mapping this model onto a Base-3 Logarithmic Spiral plane and interpreting the space as a vast "golf green", the Collatz trajectory can be viewed as a putter shot aiming for infinitely many cups scattered across the green. This framework provides an intuitive and visually engaging way to explore the structure of this integer dynamical system.

2. The Base-3 Green

Any positive integer n is mapped into the following Base-3 logarithmic polar coordinate system (r, θ):

$$r = \log_3(n)$$

$$\theta = 2\pi \cdot \{\log_3(n)\}$$

In this geometric space, powers of two 2^N appear as singular points corresponding to "cups" (jackpots) into which the ball eventually falls. Because $\log_3(2)$ is irrational, the fractional parts $\{N \log_3(2)\}$ are dense in the unit interval. Consequently, these cups are infinitely and densely distributed across all angular directions of the green. This implies that no matter which direction the trajectory travels, it will inevitably come arbitrarily close to some cup.

3. Putter Shot Dynamics

Each step of the Shiftless Collatz model decomposes naturally into two components:

- **Ball Speed ($3n$):** A pure radial thrust that increases the radius r by approximately $+1$. This corresponds to the forward momentum imparted by a strong putt.
- **Grain of the Green (LSB):** A subtle lateral adjustment that slightly changes the phase θ. This acts like the grain or slope of a golf green, gently curving the ball's path. The angular change (angular velocity ω) is approximated by:

$$\omega \approx 2\pi \cdot \log_3\left(1 + \frac{\mathrm{LSB}}{3n}\right)$$

When $\mathrm{LSB}/(3n)$ is extremely small, ω becomes nearly zero, and the trajectory proceeds almost perfectly straight.

4. Trajectory Observation

Visualizing the trajectory for the seed $n = 27$—which reaches 2^{70} in 111 steps—reveals two characteristic phases.

- **High-Altitude Cruise (The Silent Straight):** During the early and middle stages, n grows exponentially, leading to the approximation:

$$\frac{\mathrm{LSB}}{3n} \approx 0$$

  Thus, angular change is negligible, and the ball travels in a long, quiet, nearly straight line, unaffected by the grain.
- **The Final Break Near the Cup:** As the trajectory approaches the phase of the cup 2^{70}, the relative influence of the LSB increases, causing a sharp rise in angular velocity. The previously straight path bends abruptly at the lip of the cup, guiding the ball smoothly into the hole. This "sharp break" is not due to a large absolute angular velocity, but rather a relative spike compared to the near-zero angular motion just before it.

5. Conclusion

The Shiftless Collatz trajectory is not a random walk. Instead, it exhibits a distinctive putter-shot dynamic:

"The ball travels with strong forward momentum, ignoring the grain until the moment its phase aligns with one of the infinitely scattered cups, where the grain induces a sharp break that guides it into the hole."

This report visualizes this characteristic behavior and provides a geometric framework for intuitively understanding the dynamics of the Shiftless Collatz model.

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